

An Exploration of Learned Values Through Lived Experiences to Design for BIPOC Students' Flourishing

Laveda Chan*
Northeastern University
Boston, MA, USA
chan.lav@northeastern.edu

Dilruba Showkat*
Northeastern University
Boston, MA, USA
showkat.d@northeastern.edu

Alexandra To
Northeastern University
Boston, MA, USA
a.to@northeastern.edu

ABSTRACT

Prior research has primarily focused on the negative experiences faced by BIPOC students and sought to identify ways to counter harms. In contrast, our work seeks to characterize and support the positive, valuable, and meaningful experiences of BIPOC to support their holistic thriving at predominantly white institutions (PWIs). Experiences where students can share their stories and engage with racial identity development, are paramount to their flourishing. However, these stories and connections are not widely accessible. In this work, we gathered stories of meaningful experiences by conducting a qualitative semi-structured interview with 17 BIPOC students from diverse races and ethnic backgrounds, all studying across various PWIs. Our preliminary findings revealed that students derive meaningful experiences by engaging in activities that lead to the cultivation of learned values and personal growth. We discuss and situate our findings within a positive design framework to support BIPOC students' sustained well-being and flourishing.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**; **Empirical studies in HCI**; **Computer supported cooperative work**.

KEYWORDS

BIPOC, Predominantly White Institution (PWI), meaningful experiences, CHI, CSCW

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1 BACKGROUND AND MOTIVATION

Black, Indigenous, People of Color (BIPOC)¹ students at Predominantly White Institutions (PWIs) experience a range of racialized

experiences from overt racism including hostility, discrimination, and exclusion [11, 27, 43] to more subtle microaggressions including exclusions and slights to positive racial identity development and community engagement. The majority of research focused on BIPOC students narrows in on how racial harm impacts physiological, psychological, and academic well-being [20, 21] or seeks to address structural barriers, achievement gaps, and retention at individual, institutional, and community levels [14, 25, 34, 43]. While this work is important and necessary, these are all characterized by as a deficits-based approach.

Fewer studies explore how BIPOC students thrive, socialize, and engage with their racial/cultural identities in PWIs [1, 13, 15, 17, 24]. Positive aspects, such as participation in extracurricular activities, living on campus, and meaningful conversations, can function as sources of strength and have been linked to persistence and satisfaction throughout students' enrollment in the college [15]. Recent works in Human-Computer Interaction (HCI) have started to address technology design interventions that support BIPOC generally and have emphasized the significance of critical engagement with race [4, 22, 31–33, 36–38, 40, 42]. In *Designing for BIPOC Flourishing*, To and colleagues argue that the sole focus on deficits can ultimately harmfully cement a perspective in research and design that BIPOC are in a permanently lower status [39]. We see an opportunity to support BIPOC students at PWIs by uplifting their stories of flourishing and subjective well-being.

In HCI and design, several prior works have examined design for well-being on a broader scale [8, 12, 29, 41]. Desmet and Pohlmeier [9] introduced a positive design framework comprising three key elements: pleasure, personal significance, and virtue. The framework explicitly aims to foster human flourishing by emphasizing the subjective well-being of both individuals and communities.² Subsequent studies have further developed this framework tailored to specific contexts. For instance, Dennis proposed a theoretical framework addressing online well-being during the pandemic [8]. In a recent study, positive design principles were applied to analyze determinants, activities, and outcomes that contribute to sustained subjective well-being in a specific product, such as Headspace [44].

Inspired by all of the above, our work aims to integrate knowledge from social science literature into the HCI discipline to uncover technology design interventions that support BIPOC students flourishing. We transition our focus from a deficit lens to a positive design framework, aiming to comprehend and enhance the

^{*}Both authors contributed equally to this research.

¹<https://www.cbsnews.com/news/bipoc-meaning-where-does-it-come-from-2020-04-02/>

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²Subjective well-being has been defined in various ways. For example, it is defined as "the experience of joy, contentment or positive well-being, combined with a sense that one's life is good, meaningful and worthwhile" [19, p.32]. In positive design, the terms subjective well-being and happiness are used interchangeably [9]. Positive design explicitly focuses on human flourishing through design.

strengths and positive factors prevalent within BIPOC college students [45]. In doing so, we resist stereotypes that claim BIPOC students lack the capital necessary for social and professional development [1, p.5] or have cultural deficiencies that schools perceive as needing to be fixed [6]. Liane Hypolite found that one of the key mechanisms that make racial cultural centers so important and successful at supporting BIPOC students is by providing a space to share meaningful stories about their racialized experiences [13]. By collecting and sharing meaningful BIPOC student experiences, we can identify positive values that are conducive to their thriving and to disseminate these insights within the HCI community.

The objective of our study is twofold: (1) understanding the positive and meaningful experiences of BIPOC students attending PWI and (2) leveraging those stories in a positive design framework that cultivates students' sustained subjective well-being and flourishing. In that regard, we conducted qualitative interviews with 17 BIPOC students, revealing enriched experiences grounded in social and personal resources and values. Specifically, we sought to answer the research question: *what are the various positive, valuable, and meaningful PWI experiences that come from being a BIPOC student at PWIs?* This work is still in progress, but we present preliminary findings to open further conversations on designing holistically for BIPOC students. Our contributions include an in-depth analysis of learned values obtained through lived experiences and situating those values and positive experiences within a positive design framework to enable BIPOC students' subjective well-being and flourishing. In this work, we analyzed the learned values through the lens of Friedman et al. [10]'s definition of value as "*what a person or group of people consider important in life*" [10, p.2], * [1]'s interpretation of culture as "*behaviors and values that are learned, shared and exhibited by a group of people*" [1, p.7], and Le Dantec et al. [16]'s formulation of values as lived experience [2], that is, "*the values that shape and are shaped by that lived experience*" [16, p.1147].

2 METHODS

2.1 Recruitment

We sought a comprehensive understanding of BIPOC students' experiences at PWIs by recruiting participants from various institutions through posted flyers and social media. 17 students and recent graduates (Median age: 22 years; 14 female, 2 male, 1 genderqueer) from 14 private and 3 public universities in the American East-Northeast, and West regions, respectively, completed the study. All universities were doctoral or R1 research institutions enrolling both graduate and undergraduate students. Racially, of 17 participants, 5 were self-identified as Asian-American, 6 were Black, and 6 were Asian. Participants received \$20 compensation for their time and participation in this study. The study was approved by the Northeastern University Institutional Review Board (IRB).

2.2 Interview Study Protocol

We employed a semi-structured study protocol conducted remotely due to COVID-19, using Zoom for hour-long interviews conducted in English. One researcher posed questions, and a second researcher took notes during video-recorded interviews, and audio data were used for analysis. Two rounds of interviews occurred from October

2021 to April 2022. The protocol included open-ended questions about majors and favorite places on campus, followed by specific inquiries about social time, friendships, and classroom experiences to glean a holistic understanding of both positive and negative aspects of BIPOC students' college experience. Lastly, participants were inquired about their memorable and worthwhile experiences during their attendance at PWIs.

2.3 Data Analysis

We gathered 17+ hours of audio data and 100+ pages of notes, exporting MP3 audio files and initial transcripts from Zoom. Three researchers cleaned transcripts, utilizing Atlas.ti for qualitative coding.³ Our iterative analysis, incorporating inductive and deductive coding, involved generating deductive codes from research questions, prior research, and interview scripts. We followed thematic analysis (TA), starting with open codes, followed by axial coding, and then selective and affinity diagramming coding [3]. We also applied in-vivo codes, i.e., direct excerpts/quotes forming codes when appropriate [7]. Researchers independently coded all data, discussing and finalizing themes collaboratively, ensuring congruence before reporting major findings.

2.4 Researcher Positionality and Self Disclosure

Previous work by Brooks et al. [4] discussed the potential of illuminating strengths with which BIPOC currently thrives through story narrative. Informed by this and based on our own and peers' frustrations being pigeonholed into a deficits-framework, racialized experiences, and to resist normativity and bring about social change, our research team was motivated to discover the positive aspects and meaningful experiences among BIPOC students at PWIs. Our wider research team consisted of academics, students, and industry professionals having diverse racial backgrounds of Asian-American, Black, and Asian. We have diverse genders, sexuality, and originated in different parts of the world - informing our specific and diverse views on race and ethnicity. Currently, we all are based in the US. Our positioning, experiences, and identity shape our beliefs, epistemologies, and our approach to research. The goal of this research was to bring BIPOC students' meaningful experiences to the fore – as a form of resistance against systemic oppression and inequalities [18, 39].

3 FINDINGS

3.1 Learned Personal Values

The emergent values that BIPOC students perceived as essential to their flourishing at PWI include resistance to adversity, aspiration for their futures, gratitude towards support, and pride in their achievements and identity-relevant growth. These nascent values, whether applicable to individual experiences or framed as community-oriented, can elucidate how BIPOC students learn, develop, and ultimately flourish at PWIs. These values, although discussed separately, were highly interconnected, forming a mindset and interpersonal orientation that guided BIPOC students in navigating their PWI experiences.

³All authors performed data analysis manually without any advanced Atlas.ti functionalities.

3.1.1 Resistance. Students demonstrated resistance through three key actions: awakening to racial systems, PWI resistance, and active resistance. Exposure to systemic, racialized experiences differed between domestic and international BIPOC students. Some participants who were raised domestically acquired knowledge of racial issues in high schools, while others through taking relevant coursework. For instance, P001 described their race education via course work were “*super formative*”; on the contrary, P017, described that their race education was “*unproductive*” because the course was mostly taught from a “*white-centered perspective*” and the course presented a false illusion of racist experiences through colorblindness⁴, P017 said that, “*at least [in] my experience that racism was kind of a thing of the past.*” For international students, awareness of racial issues was a result of experiences attending PWI. P038, an international Black minority, attributed her awakening to being othered: “*I think some people try to make connections to connect me to some form of the life that they’ve had [...] like you are other, and then we try to find the other in my life, so I connect to the two others. I mean now you kind of get used to it because it happens so much*” (emphasis original). This awareness was framed as a form of resistance because it allowed them to further develop other skills while challenging the dominant narratives that often encouraged its constituents to ignore race and associated issues.

While this research focuses on the positive experiences of BIPOC students, they coexist alongside experiences with interpersonal, institutionalized, and ideological racism [11, 27, 43]. As such, consistent with prior work, our participants similarly faced racial, cultural, and institutional barriers within and outside the classroom [20, 23]. For example, participants shared how professors exhibited microaggressions when they joked about mispronouncing “*difficult*” names. Black students expressed experiencing increased pressure when they were the sole Black students in core curriculum classes. When singled out, they felt a sense of invisibility or the burden of representing the entire Black student population. While some students resisted these microaggressions by “*just getting used to it*” and choosing not to let these moments infringe on their sense of identity, other participants resisted by altering viewpoints and engaging in activism. For example, P019, who attended a predominantly white school prior to college, learned to navigate such spaces with “*skill sets: like don’t get into arguments with people [...] do what I need to do, and not [put up with] those comments*” and most importantly, “*show them we’re just as capable as [they] are.*” On the other hand, P004 gravitated towards engaging in organizing student-led groups that focused on addressing race-related recruitment and funding disparity, through which she found purpose and community, “*I have a lot of friends from that program, [...] but I think, [that’s] what kept me going.*”

In sum, despite facing racial challenges, BIPOC students exercised resistance, thus transforming these negative experiences (similar to joy as a form of resistance [18]) into positive outcomes and opportunities for personal and professional growth and skill development.

3.1.2 Aspiration and Hope. While attending PWI, BIPOC students developed ideological (personal) and vocational (professional) goals

that reflected their hopes and aspirations for a better future, both for themselves and the community. While some participants focused on achieving personal well-being and happiness, others aimed to address racial disparities in future professions. For example, P046 envisioned achieving success without compromising inner peace and happiness. They described, “*I’m an emotional person, so that’s where the happiness comes from. So if I got a job offer from [anonymized] right now [...] and I worked there for years. If there’s a time when I feel unhappy because everybody’s thinking and you’re at the top, [...] then I would like to find something else. I want to be successful in my own right but also happy.*” Another participant, P054, echoed a similar desire: “*one personal goal of mine is just to be personally, mentally, physically the best that I can, meaning that I feel like I’m able to thrive in the environment.*”

Other participants expressed aspirations to pursue professions that directly address racial disparity. P017 hopes to pursue a graduate degree and a career in cognitive-behavioral psychology motivated by the lack of Black practitioners in this field. She further mentioned, “*I want to pursue getting my Ph.D. I want to go into psychotherapy because only 4% of all therapists in the United States are Black [...] One facet of my career will be dedicated to having a predominantly Black or predominantly LGBTQ+ or POC client base.*” In similar cases, participants set more immediate goals. P004 utilized her previous DEI advocacy experience from college in her current role at work to continue working on racial justice: “*I’ve consistently cared about people and wanted to create a space for people [...] [that] has been like my early career focus, like advocacy [...] racial justice is [...] being able to create spaces, where especially Black and indigenous folks are at the center of conversations and are protected and are thriving.*”

That said, BIPOC participants describe their aspirations and hopes as a significant part of their experiences at PWIs. While their goals may be personal or community-oriented, they collectively demonstrate a determination to forge meaningful futures, simultaneously resisting racist climate surrounding them.

3.1.3 Pride. As previously described, BIPOC students frequently resist experiences of interpersonal, institutionalized, and ideological racism. As a result, participants took pride in their academic achievements, community contributions, and personal growth. Many participants highlighted their determination and hard work, sharing stories of completing degrees or thriving in challenging academic environments. For example, P004 finished both undergraduate and graduate degrees without financial support from family, a feat that filled her with pride: “*I have a degree from [anonymized R1 school], those things are huge and they’ve definitely given me so many opportunities coming out of those two experiences that I didn’t imagine.*” P007 on the other hand was proud of what she was able to do even without a degree in healthcare disparity education by leading the youth and training them to become confident leaders: “*we are made up of college students college students and my mentor who is just a medical student [...] with no credentials [...] what we’ve been able to do the community definitely brings me a lot of joy.*”

Several participants also expressed that adapting to cultural changes, being able to share their culture, and learning how to be independent were key aspects that made them proud. P054, an international student, shared that a source of pride was “*being more*

⁴<https://www.learningforjustice.org/magazine/fall-2009/colorblindness-the-new-racism>

independent, being able to not only gain an education here but also being able to work towards caring for myself as an individual— so being able to manage cooking for myself or looking over [the] apartment that I am living in and just being able to [manage] those responsibilities.” BIPOC students also actively engaged in the coordination and execution of school-sanctioned cultural events, participating in the managing, organizing, and fund collection process. One example comes from P029 who organized their cultural new year celebration, and shared their emotional response to having implemented such an event: *“it was a great success, [...], next year, we got more funding and we [will] make the show even bigger. We had a reputation of [a] cultural organization that put on a very good show [...] and that’s something I’m very proud of.”*

Upon reflection, BIPOC students shared their personal and professional accomplishments, things they were able to do during their time at PWI despite being racial minorities, attaining degrees, getting jobs, serving the communities, helping peers and friends, and seeing them thrive. All of these contributed to BIPOC student’s source of pride, while they simultaneously possessed deep awareness of color blindness and actively resisted systemic oppression.

3.1.4 Gratitude. Participants were thankful for positive opportunities, the support they received, meaningful connections, and personal development they experienced at PWIs. Some relied on faith practices as a source of gratitude that helped them navigate challenges and maintain a positive outlook despite the systemic issues at PWIs [26]. P017 shared that while she was not necessarily “**desensitized**” to systemic racial issues, she would “*rather focus on the good experiences my Blackness brings upon me, like social events and all of my friends and just being able to participate in my culture freely which is a beautiful thing.*” P004 similarly expressed gratitude for her support network, sharing “*I also don’t take it for granted, so that’s a lot of support and a lot of people around me who have given me these opportunities and believed in me.*” Similarly, another participant, P024 expressed gratitude at her opportunity to do interdisciplinary work, merging technology and biological sciences, and to be able to work with a supportive mentor who guides her: “*I’m grateful to have this opportunity [...] I love the people at my company [they are] so nice like my manager [...] I’m so glad I’m able to have such a close bond with my manager. [...] I’m just so happy there.*”

These examples underscore the gratitude expressed by BIPOC participants towards their support network that helped them navigate through adversity, especially challenges posed by the racist climate at PWIs.

3.2 Actively Applying Learned Values into Practice

Our research underscores the proactive efforts of BIPOC students in translating their learned values into actions— challenging their negative experiences by supporting peers and students with shared identities. Students were able to identify the skills, knowledge, resources, and tools in order to positively engage peers by creating additional safe spaces and opportunities and developing interpersonal skills. For example, P058 planned and executed an inaugural African Student Dinner in collaboration with the African Student Organization prior to Thanksgiving Holidays. This event was catered towards international students, especially those who were

from Africa, and often could not travel home for Thanksgiving and Christmas holidays. P058 said, “*It fills me with so much joy because of the kind of feedback we got,*” as the event garnered 800+ students in attendance who experienced joy and were able to interact and relate to another without “*having to be a warrior of the continent or require a fake accent.*”

P058 went on to explain the significance of not only physical spaces for connection but also spaces for resource sharing. They shared, “*we don’t tend to get assistance from the university when we want to do things [...] to pursue our goals the same way other people can.*” To subvert this experience, P058 partnered with an alumni organization that was able to raise funds for students from the African continent. This initiative enabled peers within their community to pursue the same experiential learning opportunities without being hindered by harsh cost-related implications: “*we’ve been able to get a fund for African students to get some form of stipend if they are going through some financial difficulties in terms of paying their fees, or if they just want to experience the world. Because we should be allowed to [despite] our nationality.*”

Participants stressed the importance of developing effective communication skills that contributed to their professional development. As students had to overcome obstacles unique to their race and identity both within academic classrooms and beyond, the cultivation of interpersonal skills and the consequent growth evoked values of pride and gratitude. P027 described struggling to interact with other students from different ethnic backgrounds during the first two years of college. Through prioritizing the intricacies of cross-cultural communication, P027 was able to develop relationships and friendships in the later years of college. In their words, “*it’s been a character development for the past four years.*” To resist the lack of diversity at their PWI, P027 rushed a frat: “*I was grateful to meet a lot [of] people through my frat and through many other friends. And that was the biggest joy— that there’s someone who wants to understand my culture [and] how I approach life.*”

More than fostering friendships, the development of interpersonal skills allowed P027 to grow professionally. P027 added, “*I got my job essentially through communicating and networking [...] I definitely owe my freshman year tragedy to this development, and then in second year the development of Greek life or my involvement in Greek life. All of that contributed to [...] one of [my] biggest strengths today.*” Importantly, both P058 and P027s experiences exemplify the applied value of resistance. Their decisions to seize opportunities for community and personal growth ultimately countered and resisted racial barriers experienced by minority communities, thus underpinning the significance of centering positive experiences and associated values to uplift BIPOC students during their attendance at PWIs.

4 DISCUSSION

4.1 Situating Findings within the Positive Design Framework for BIPOC Students’ Flourishing

In the following, we situate our study findings within the positive design framework proposed by Desmet and Pohlmeier [9] to provide technical design intervention for supporting BIPOC students’ sustained subjective well-being (as opposed to objective well-being

such as housing or nutrition). This framework comprises three key ingredients (with overlapping attributes [28]) that are essential to sustaining a community and an individual's well-being: pleasure, personal significance, and virtue. While designing for well-being can only focus on one key aspect, it should avoid adversely affecting the other two. Conversely, in implementing positive design or designing for flourishing, equal attention should be dedicated to all three ingredients, ensuring a balanced approach [9, 29]. We situate the identified learned values within the third ingredient virtue as they guided BIPOC students through their lived experiences at PWI and encouraged moral action. The application of their learned values, i.e. the actions guided by their values, can be dispersed across all three components, as they are actively influencing and being influenced by their lived experiences and values. Intentional and positive design incorporating BIPOC students' meaningful experiences, activities, and acquired values discussed across each of the positive design ingredients in the design process is critical, since the efforts, intellectual and informal interpersonal activities heavily influence BIPOC student's personal growth and flourishing [9, 15, 39].

Designing for pleasure constitutes *"a person's momentary pleasures"* [9, p.4], attained by embracing the moment in the absence of stress, and experiencing relaxation and positive emotions. Interview data revealed that BIPOC students expressed such sentiments when discussing the activities they engaged in within their preferred places on campus, including sitting quietly in a university hall and socializing at cultural clubs. These activities derived feelings of joy, comfort, and pleasure because they were able to reflect, share, and care for their peers, a moment of respite from their busy academic schedules and multiple demanding responsibilities. As such, actions enabled by applying learned values, such as P058's organization of a holiday party, created a space where BIPOC students can freely reflect and socialize— exemplifying the coexistence of applied values and pleasure in BIPOC students' experiences at PWIs. As our findings suggest, pleasure was one aspect of students' PWI journey and incomplete without future wishes (e.g., anticipation [5]), achievement, and values [15], and a desire for systemic change through personal and professional goals and aspirations.

Personal significance, characterized as *"happiness that comes from a sense of personal meaning"* [9, p.5]. In our research, BIPOC students desired to attain academic success, a job [19], create life-long friendships, communities, and connections [28, 30], enjoy new hobbies through which they can find comfort and attain self-growth [35], and hone problem-solving skills when faced with novel situations [15]. Similarly, Section 3 describes how BIPOC students' learned values reflect across their attendance at PWIs but also beyond. Students established both long-term and short-term goals and participated in activities that both enabled personal happiness and well-being and bridged resource disparity in minority communities— eliciting feelings of pride, and gratitude, but also fulfillment. Consideration should be given to these actions and activities, that is the association of intersectional identities with personal goals, and their positive impact on BIPOC students, when **designing for personal significance**.

Further, section 3 identifies and describes four values that were perceived as significant for BIPOC students: resistance, aspiration and hope, pride, and gratitude. As previously elaborated, these

values virtuously guided and influenced BIPOC students' decisions and actions at their PWIs and also in life in general. In accordance with prior research, they acted virtuously by being open-minded and appreciating diversity [15], celebrating each other's success, and committing to personal goals [44]. As such, these values should be applied when **designing for virtue**, as they foster growth, promote sustainability [28], and contribute to BIPOC's flourishing [39]. Designing technologies for underserved BIPOC students is essential to enacting countercultural resistance, social justice, and most importantly establishing – joy as a source of resistance for coping, healing, and thriving [9, 18, 39].

4.2 Ethical Considerations

Adhering to the human values of resistance, pride, hope, and gratitude has positive outcomes for BIPO students. However, designing for sustainable subjective well-being using the positive design is not free from its ethical debt due to its persuasive abilities. Likewise, it is important to practice awareness and give special consideration to the potential negative and societal implications of technology design, by focusing on balancing the key ingredients of positive design. Future technologies designs guided by values have the potential to empower BIPOC students through engaging them in activities (e.g., long-term goal, individual fit), without trying to alter their behavior in any way because the end goal of positive design is to support flourishing – *"flourish without being prescriptive"* [9, p.11].

4.3 Limitations

We acknowledge that BIPOC students have diverse experiences, not only across different racial groups but also intersecting with other identities such as citizenship, gender, etc. Similar to any other explorative qualitative study our study findings have several limitations. Some limitations are directly related to the generalizability of the findings. Our participants were mostly STEM majors, five (29%) of them were international students, while the rest (70%) of them were originally from the U.S. 14 (83%) of the participants self-identified as women, and only two (11%) of them self-identified as male. Almost all of the participants 14 (83%) self-reported being studying at private US colleges, and only 3 (17%) of the students of color were from public institutions. Collectively, all of these can possibly have contributed to the findings reported in this study.

5 CONCLUSION

In this paper, we identified four positive values that were integral to BIPOC students' subjective well-being during their enrollment at PWIs: *resistance, aspiration and hope, pride, and gratitude*. Students' engagement in activities by applying learned values enabled them to achieve professional success such as finding jobs and internship opportunities, interpersonal success from establishing friendships and cultivating community, and importantly, personal happiness in an environment where racist experiences are pervasive. We situate these positive values and their applications across the three main ingredients of a positive design framework: pleasure, personal significance, and virtue. When designing for BIPOC students, we emphasize the need for HCI designers and researchers to be intentional about integrating positive values and activities into the design process. Through our findings, we believe in the potential

of technology or product design to explicitly nudge students into activities before, during, or even after their attendance at PWIs – to cultivate personal growth, positive development, subjective well-being, and flourishing. Lastly, in our future work, we plan to build on some of the stories in-game narrative storytelling due to stories' inherent power of connecting people (e.g., to feel seen, included) with stories who share similar experiences [13]; in this process, we also hope to place our adapted positive design framework into action.

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